Rajasthan **Board Class**

Subject – Physics

Time: $3\frac{1}{4}\,Hours$ **M.M: 56**

1. **Calculate electric potential at a point 1m distance from a point charge of** 10^{-9} **coulomb.**

   Sol:

   $$V=\frac{kq}{R}$$
   $$=\frac{9\times10^{9}\times10^{-9}}{1}$$
   $$=9V$$

2. **In given diagram value of carbon resister is** $22\times10^{5}\Omega\pm5\%$ **.Write colour of first ring A.**

   Sol:

   Colour of ring:

   0 1 2 3 4

   B B R O Y

3. **Write formula for force on a current carrying conductor in a magnetic field.**

   Sol:
   $$F=\pm lB\sin\theta$$

4. **Define angle of dip. Write value of angle of dip at magnetic poles of earth.**
   Sol:
   $$\theta=90^{\circ}$$

5. **Write relation between route mean square (rms) value and peak value of alternating current.**
   Sol:

$$I_{rms} = \frac{I_o}{\sqrt{2}}$$

6. **In LCR alternating circuit,** $R = 10\Omega,\ X_L = 100\Omega$ and $X_C = 100\Omega$**.Write value of impedance of circuit.**

   Sol:

   $$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
   $$Z = R = 10\Omega$$

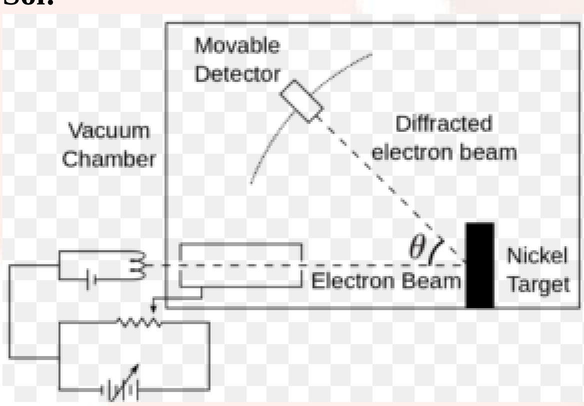
7. **Write relation between power of lens and it's focal length.**

   Sol: $P = \frac{1}{f}$

8. **Define threshold frequency.**
   Sol:

   The minimum frequency of the incident light which can cause photo electric emission that is known as threshold frequency.

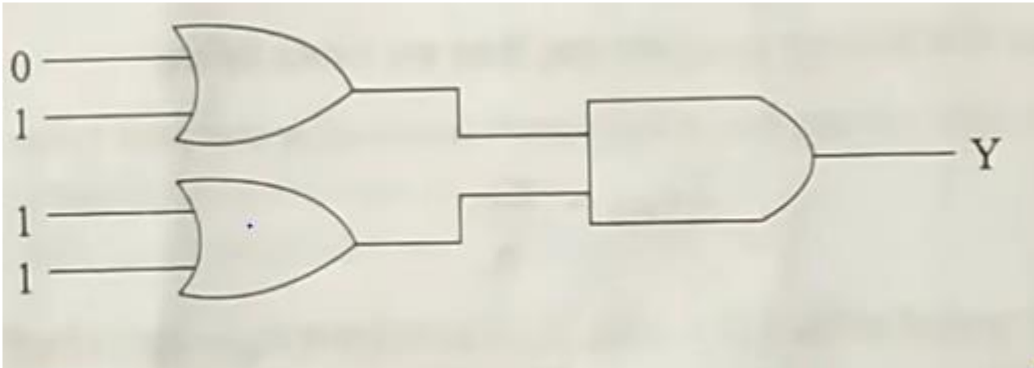
9. **Draw diagram of the experimental arrangement of davission and germer experiment**
   Sol:



10. **Write name of series of hydrogen spectrum obtained when an electron transist from higher energy level** $n_2 = 2, 3, 4, 5.........$ **to ground energy level** $n_1 = 1$**.**

    Sol:
    Lyman series.

11. **Write value of output Y in diagram.**



Sol:

Input of first OR gate $1+1=1$

And, Input of second OR gate $1+0=1$

Now, Final input for AND gate is $1 \cdot 1=1$

12. Define modulation.

Sol:

It is a process through which audio, video, image or text information is added to an electrical or optical carrier signal to be transmitted over a telecommunication or electronic medium.

13. In electromagnetic waves, write the value of (A) angle and (B) phase difference, between electric field $\vec{E}$ and magnetic field $\vec{B}$.

Sol:

Angle is 90°.

And Difference between,

$E = E_o \sin(kx - \omega t)$

$B = B_o \sin(kx - \omega t)$

Then,

14. Define

a) Electric dipole moment.

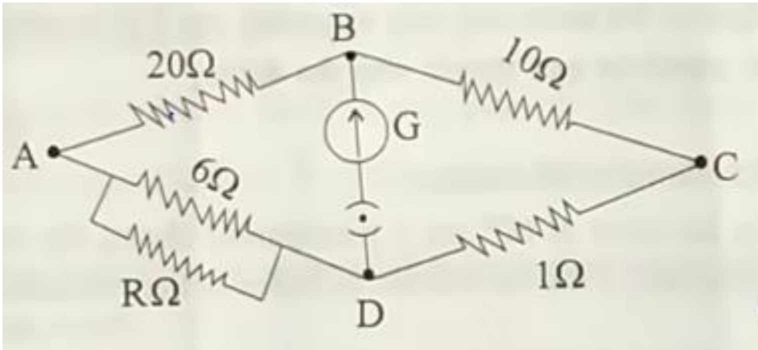
b) Equipotential surface.

Sol:

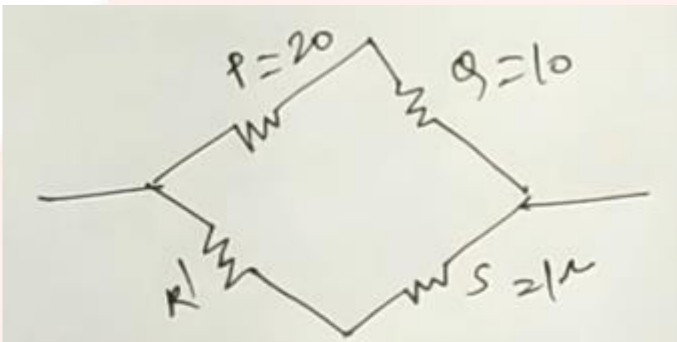
a) A measure of the separation of positive and negative electrical charges in a system of electric charges, that is, a measure of the charge system's overall polarity that is known as electric dipole moment.

b) The surfaces of constant scalar potential is called equipotential surface.

15. Calculate the value of unknown resistance R in given circuit. If Wheatstone bridge is in balanced condition.



Sol:



For parallel:

$$\frac{1}{R'} = \frac{1}{6} + \frac{1}{R}$$
$$\frac{1}{R'} = \frac{R+6}{6R}$$
$$R' = \frac{6R}{R+6}$$

And then,

$$\frac{P}{Q} = \frac{R'}{S}$$
$$\frac{20}{10} = \frac{6R}{(R+6)\times 1}$$
$$20R + 120 = 60R$$
$$120 = 40R$$
$$R = 3\Omega$$

16. (a) Define Curie temperature.

(b) The pole strength of poles of a bar magnet of effective length $0.1m$ is 40A-m. **Calculate it's magnetic moment.**

Sol:

(a) The temperature above which a ferromagnetic substance loses its ferromagnetism and become paramagnetic. Curie point temperature the degree of hottest and coldness of a body or environment based on word net 3.0 , farlex and clipart collection.

(b) We know that,

The magnetic moment of the bar magnet is,

$M = m \times 2l$

$= 40 \times 0.1m$

$= 4Am^2$

17. Write Lenz's law. Lenz's law obeys law of conversation of energy. Explain.

Sol:

The induced current produced in a circuit always flows in such a way that it opposes the change or the cause that produces it is known as Lenz's law.

According to the Lenz's law, the induced emf opposes the change that produces it. It is this opposition against which we perform mechanical work in causing in the magnetic flux.

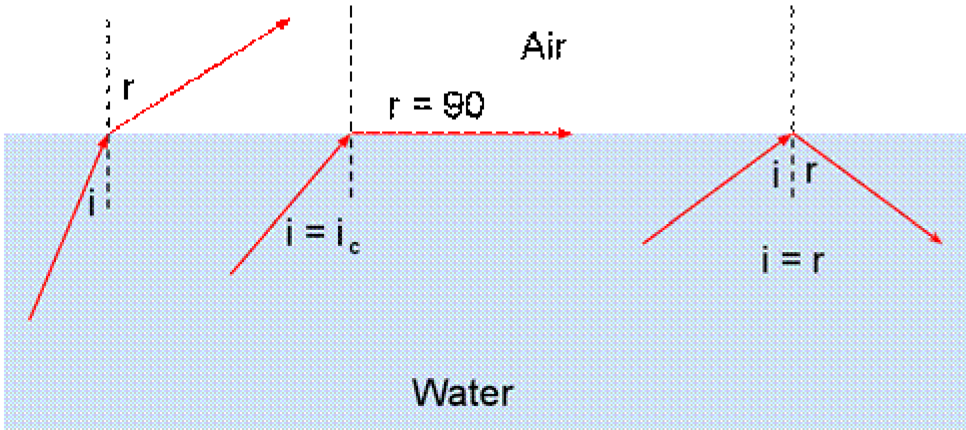
Suppose when a north pole of a magnet is pushed towards coil.Then the direction of induced emf in the coil is such the end facing the magnet becomes north pole so as to oppose the motion of the magnet. Thus work has to be done against the magnetic repulsive force to push the magnet towards the coil.The electrical energy produced in the coil is at the expense of this work done. so, the mechanical energy is converted into electric energy.

18. (a) Total internal reflection

(b)Diffraction of light.

Sol:

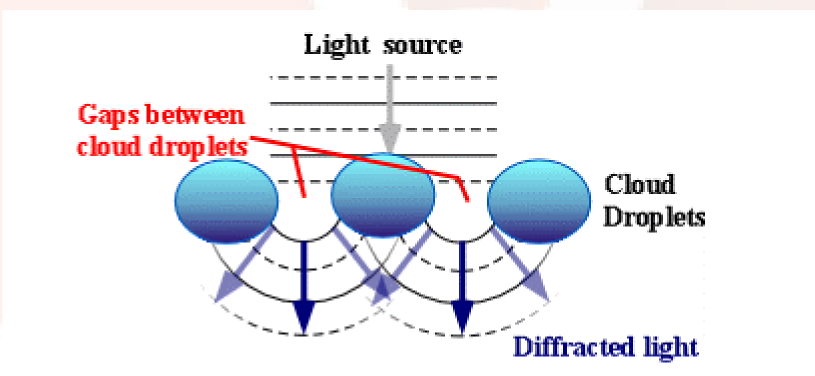
(a) When light is incident from a **denser to rarer** medium the ray deviates away from the normal. For a certain angle of incidence $(i = i_c)$ the angle of refraction becomes 90^0. This angle of incidence is called the critical angle.



This phenomenon is called Total Internal Reflection.

(b)

Diffraction is the slight bending of light as it passes around the edge of an object. The amount of bending depends on the relative size of the wavelength of light to the size of the opening. If the opening is much larger than the light's wavelength, the bending will be almost unnoticeable. However, if the two are closer in size or equal, the amount of bending is considerable, and easily seen with the naked eye.



19. (a) Write formula related to Malus law.

(b) When light is incident 60^0 **on a transparent sheet, the reflected light is completely polarized. Find the refractive index of the substance and refraction angle.**

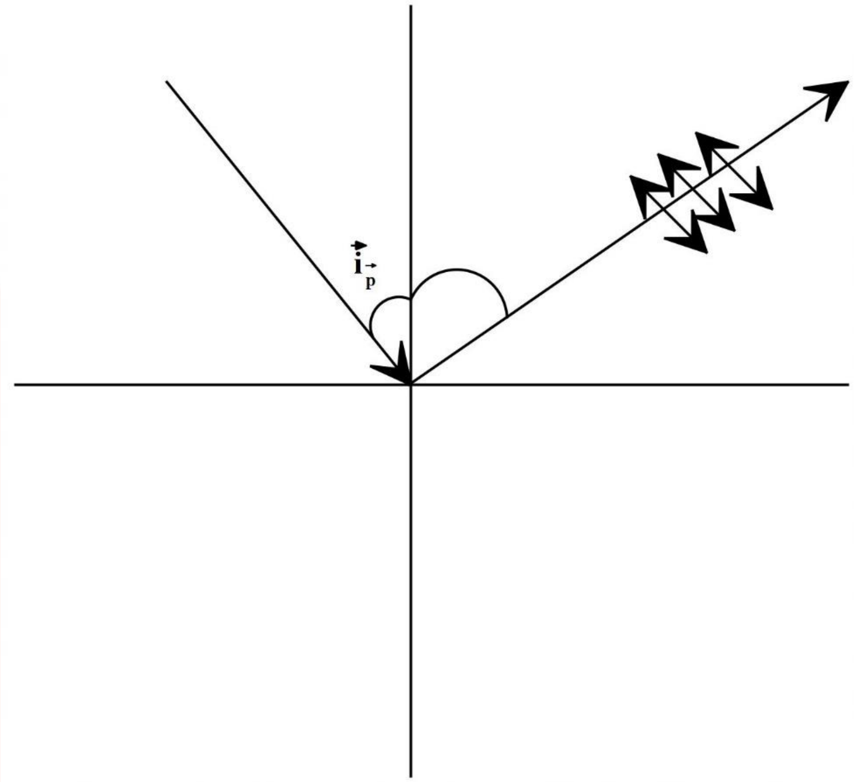
So*l:*

(a) According to Malus, when completely plane polarized light is incident on the analyzer, the intensity I of the light transmitted by the analyzer is directly proportional to the square of the cosine of angle between the transmission axes of the analyzer and the polarizer.

$I \infty \cos^2$

$\Rightarrow I = I_0 \cos^2 \theta$

(b) L



$ip + r = 90^\circ$

$lp = 60^\circ$

$n = ?$

From Brewster's law

$n = \tan ip$

$n = \tan 60^\circ$

$n = \sqrt{3}$

$60^\circ + r = 90^\circ$

$r = 90^\circ - 60^\circ$

$r = 30°$

20. Calculate de Broglie wavelength of a wave associated with an electron, which is accelerated through a potential difference of $100V$.

Sol:

$$\lambda e = \frac{12.27}{\sqrt{V}}$$

$$\lambda e = \frac{12.27}{\sqrt{100}}$$

$$\lambda e = 1.227\ \overset{o}{\mathrm{A}}$$

21. Derive formula for obtaining internal resistance of a primary cell with help of potentiometer. Draw circuit diagram.

Sol:

When key is off

$E = Kl_1 \rightarrow (1)$

and when key is on, $I \neq 0$

$V = Kl_2 \rightarrow (2)$

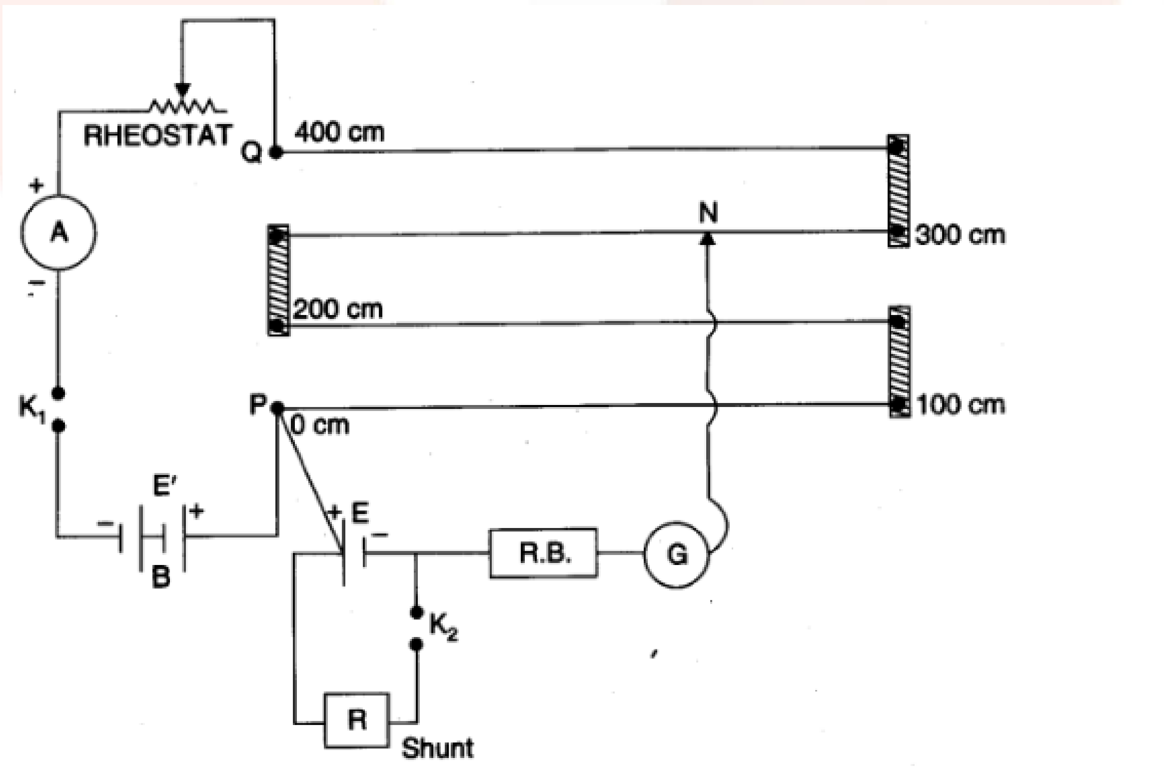
$l_2 < l_1$

bec$V < E$

$$r = \left(\frac{E}{V} - 1\right)R$$

$$r = \left(\frac{Rl_1}{Rl_2} - 1\right)R$$

$$r = \left(\frac{l_1}{l_2} - 1\right)R$$



22. Write Niles Bohr's any two postulates for hydrogen atom (hydrogen like ion)
Sol:

(1) Electrons will not radiate energy while revolving in circular orbit,, and it will only loss energy and gain so, when it make transition..
Bohr named these orbits as the stationary orbit..
(2) In the 2nd postulates Bohr explained the concept of stationary orbits..
The orbits which are the integral multiple of $\frac{h}{2\pi}$ the electrons are only allowed to revolve their and if it do so, then no energy emission takes place acc to electromagnetic wave theory.

22. Write Nieles Bohr's any two postulates for hydrogen atom (hydrogen like ions).

Sol.

1. The electrons in an atom revolve around the nucleus only in certain selected circular orbits. These orbits are associated with definite energies and are called energy shells ore energy levels. These are numbered as 1,2,3,4……etc. or designed as K,L,M,N ……etc. shells.
2. Only those orbits are permitted in which the angular momentum of the electron is a whole number multiple of $\frac{h}{2\pi}$ where h is Plank's constant. That's
   Angular momentum of the electron,
   $mvr = \frac{nh}{2\pi}$, where n= 1,2,3,…..
   Where,
   M is the mass of the electron
   V is the velocity of the electron and r is the radius of the orbit.
   In other words, angular momentum of electron is an atom is quantised.
   As long as the electron remains in a particular orbit, it does not lose or gain energy. This means that energy of an electron in a particular orbit remains constant. That is why; these orbits are also called stationary states.
3. When energy from some external source is supplied to the electron, it may jump to some higher energy level by absorbing a definite in energy between equal to the difference in energy between the two energy levels. When the electrons jumps back to the lower energy level it radiantes same amount of energy in the from of a photon of radiation.
   $\Delta E = E_2 - E_1$
   $= hv$
   $v = \frac{(E_2 - E_1)}{h}$
   Where v is the frequency of the emitted radiation.

23. (a) Secect acceptor type impurity among the following: 1+1=2

Arsenic (As), Antimony (Sb), gallium (Ga) and phosphorous (P).

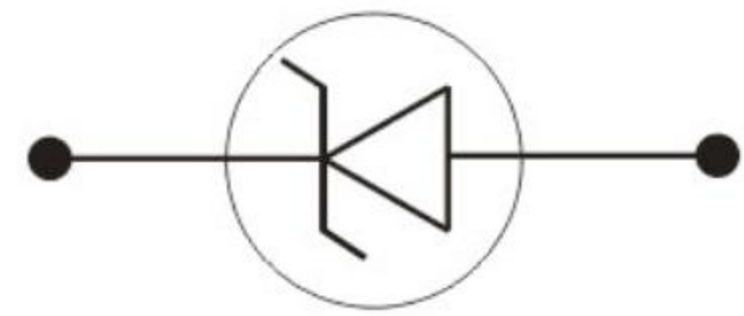
Sol.

B, AL , Ga, Sn, Tl there fore acceptor is Ga.

(b) Draw symbol of Zener diode.

Sol.

Draw the figure Zener diode below in.



Zener Diode

24. The magnitude of electric field $\vec{E}$ at a point in free space in 300 V/m. Find magnitude of magnetic field $\vec{B}$ at the point. Velocity of light is $3\times10^8 m/s$.

Sol.

$E = 300v/m$ $\quad$ $C=3\times10^8 m/s$

$B = ?$

$$C = \frac{E}{B} = \frac{300}{3\times10^8}$$

$$B = 100\times10^{-8} = 10^{-6}T$$

25. Write Rutherford-Soddy law of radioactive decay and derive related equation. Draw exponential decay curve of a radioactive substance.

Write ratio of half-life and mean life of a radioactive substance. [1+11/2+1/2=3]

Sol.

Rutherford-Soddy law- it is a self-process in the heavier nuclei.

Self-disintegration to become stable.

It is not depends on pressure, temperature, force,

It is based on the principle of probability

if

$$t = 0$$

$$-\frac{dN}{dt} \propto N$$

$$-\frac{dN}{dt} = \lambda N$$

$$\frac{dN}{N} = -\lambda dt$$

$$\int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_{N_0}^{N} dt$$

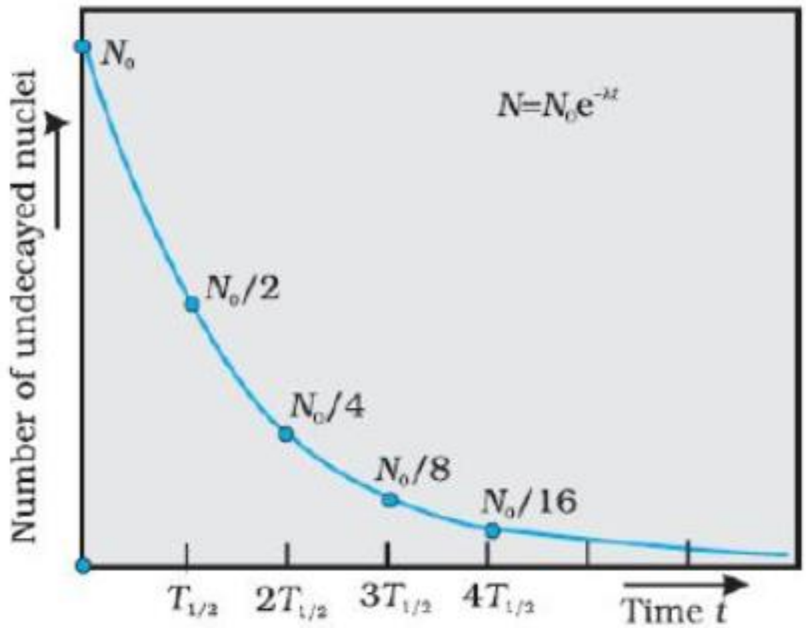
$$[\log N]_{N_0}^{N} = -\lambda t$$

$$\log N - \log N_0 = -\lambda t$$

$$\log\left(\frac{N}{N_0}\right) = -\lambda t$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$N = N_0 e^{-\lambda t}$$



Exponential decay of a radioactive species. After a lapse of $T_{1/2}$, population of given species drop by factor of 2.

Radion half-life and mean life of a radioactive substance –

$$\frac{T}{T_n} = \frac{0.693}{\lambda \times \frac{1}{\lambda}}$$

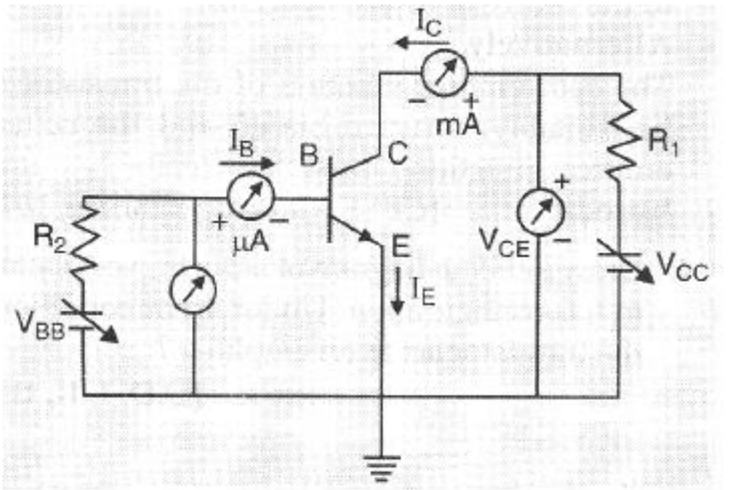
$$\frac{T}{T_n} = 0.693$$

26. Describe the experimental set up obtaining output characteristic curve of a PNP transistor in common emitter configuration with circuit diagram. Also draw the curve obtained.

Select possible value of common base current amplification factor α of transistor among the following:

0.9,9,19,49 and 99.

Sol:



npn transistor work as a amplifiar.

Its use in soundbox....radio...etc

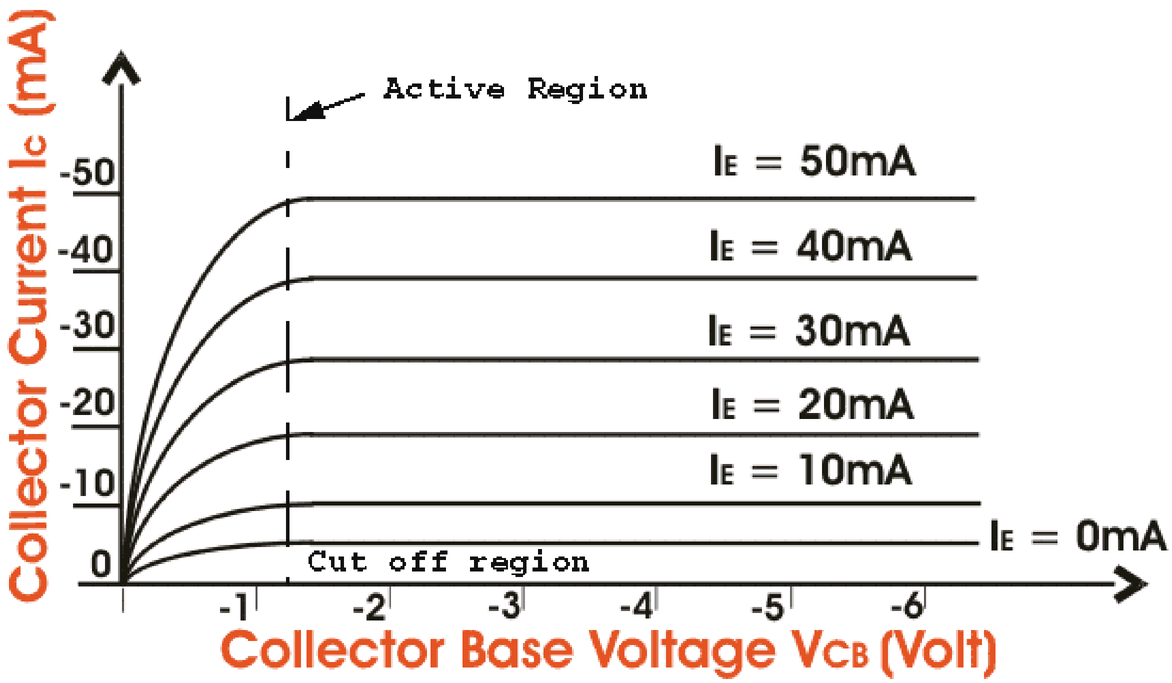
.If input current increase so output voltage increse.

At the point when input and output current go ing to reverse bias.

That's called cut of area.

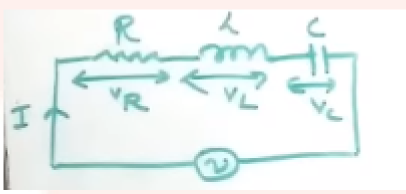
Here, the value of α is always less than 1 $\left(0.9\right)$ and the value of β is always greater than 1 $\left(9\right)$.

And the characteristics curve of the following is given below,



27. Draw vector diagram (phase diagram) for series RLC circuit which is connected with an alternating voltage source and determine the expression for impedance of the circuit.

Sol.



AC though R-L-C Circuit

$$\tan\phi = \frac{V_L - V_c}{V_R}$$

$$\phi = \tan^{-1}\left(\frac{V_L - V_c}{V_R}\right)$$

$$= \tan^{-1}\left(\frac{1X_L - 1X_c}{1R}\right)$$

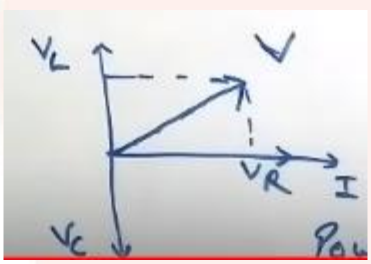
$$= \tan^{-1}\left(\frac{X_L - X_c}{R}\right)$$

$$= \tan^{-1}\left(\frac{X}{R}\right)$$

$$power\ \text{factor}\ \cos\phi = \frac{R}{L}$$

$$= \frac{R}{\sqrt{R^2 + \left(X_L - X_c\right)^2}}$$

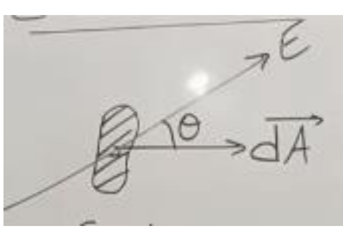
$$\text{P=VI}\cos\phi$$



OR,

28. Gauss' theorem states that the surface integral of the electric filed over the closed surface is equal to $1/\varepsilon_0$ times the charge enclosed.

Sol.



$$d\phi = EdA\cos\theta$$

$$d\phi = \vec{E}.\vec{dA}$$

$$\phi = \int d\phi = \int \vec{E}.\vec{dA}$$

$$\phi = \vec{E}.\vec{A} \to \vec{E}$$

''The net electric flux through a closed surface (3-D) is $\frac{1}{\varepsilon_0} times$

Net charge enclosed by the surface''

Gauss' Therorem

$$\phi_e = \phi_e \vec{E}.\vec{ds} = \frac{q}{\varepsilon_0}$$

$$\left|\vec{E}\right| = \frac{1}{4\pi\varepsilon_0}.\frac{q}{r^2}$$

$$\phi_e = \phi_s \vec{E}.\vec{ds}$$

$= \phi_s Ed\cos\theta$ [θ is angle between Êand dŝ]

$$=\phi_s Eds$$

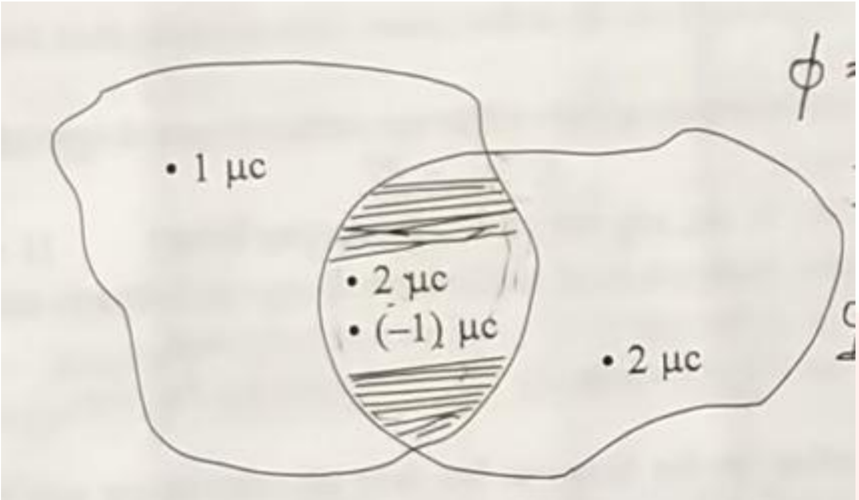
$$= E\phi_s ds$$

$$= \frac{1}{4\pi\varepsilon_0}.\frac{q}{r^2}.4\pi r^2$$

$$= \frac{q}{r^2}$$

(b) Calculate the electro flux given the result of shade region.

Sol.



$$\phi = \frac{\sum q}{\varepsilon_0}$$

$$= \frac{1\times10^{-6}}{8.85\times10^{-12}}$$

$$\phi = 1.13\times10^{+6}\,vm$$

OR,

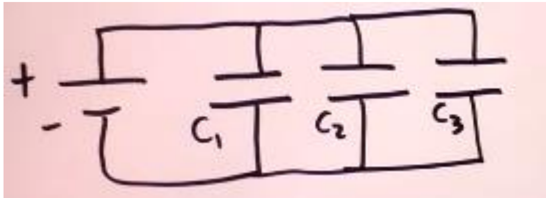
(a) Define capacitor. Draw a circuit diagram and obtain a relation for equivalent capacitance for the series combination of three capacitor.

Sol.

A capacitor is positive two- terminal electrical component used to store energy electrostatically in an electric field unlike a resistor, a capacitor does not dissipate. Instead a capacitor store in energy the form of an electrostatic field between its plates.

$V = IR$

$Q = CV \qquad V = \dfrac{Q}{c}$



$$V = V_1 = V_2 = V_3$$
$$Q = Q_1 + Q_2 + Q_3 = C_1V + C_2 + C_3V$$
$$= \left(C_1 + C_2 + C_3\right)V$$
$$= CV$$

$$V = V_1 + V_2 + V_3$$
$$= \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$$
$$= Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$
$$V = \frac{Q}{C_{eq}}$$
$$\frac{1}{C_{eq}} = \sum_{i=1}^{N} \frac{1}{c_i} = C = \frac{\varepsilon_0 A}{d}$$
$$\frac{1}{C_{eq}} = \sum_{i=1}^{N} \frac{1}{c_i}$$

(b) Find the equivalent capacitance between points A and B in the given figure.

29.

(a) Write Ampere's law.

Sol.

Ampere's law:

Any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop is known as Ampere's law.

$Finds: \vec{B}$

For symmetric situations

uses closed loops

$\mu_0 I = \oint \vec{B}.\vec{dl}$ closed path integral

$\mu_0 I\left(\text{through Amperian membrance}\right)$ $\oint \vec{B}\ .\vec{dl}$

$\oint \vec{B}$(*on* the amperian loop

$\vec{dl} = amperianloop$

(b) Draw a diagram and derive an expression for magnetic field due to an infinitely long straight current carrying conductor at any point.

Sol:

Let us consider a circular loop of radius r **around an infinitely long straight wire carrying current I. As the field lines are circular, the field vector B at any point of the circular loop is directed along the tangent to the circle at that point. By symmetry, the magnitude of field vector B is same at every point of the circular loop.**

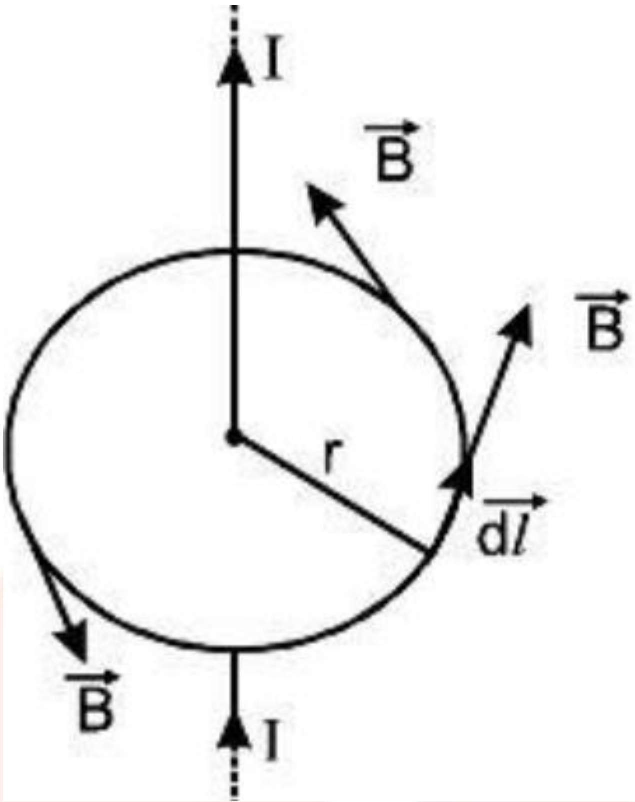
Therefore,

$$\int_b^a b\cdot dl = \int_b^a Bdl\cos\theta = B\int_b^a dl = B\cdot 2\pi r$$

From Ampere's Circuital Law

$B\cdot 2\pi r$

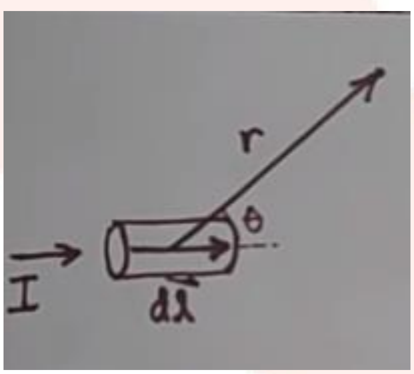
$$\therefore B = \frac{\mu_0 I}{2\pi r}$$



OR,

a) Write Biot-Savart law.

Sol.

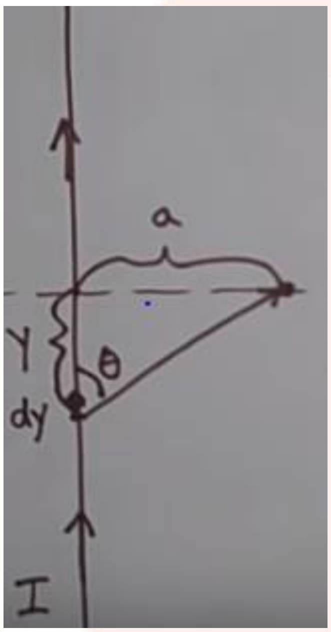


$$\overrightarrow{dB} = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$

$$\frac{\mu_0}{4\pi} = \frac{\mu_0 Idy}{4\pi\left(y^2+a^2\right)\left(y^2+a^2\right)^{1/2}}$$

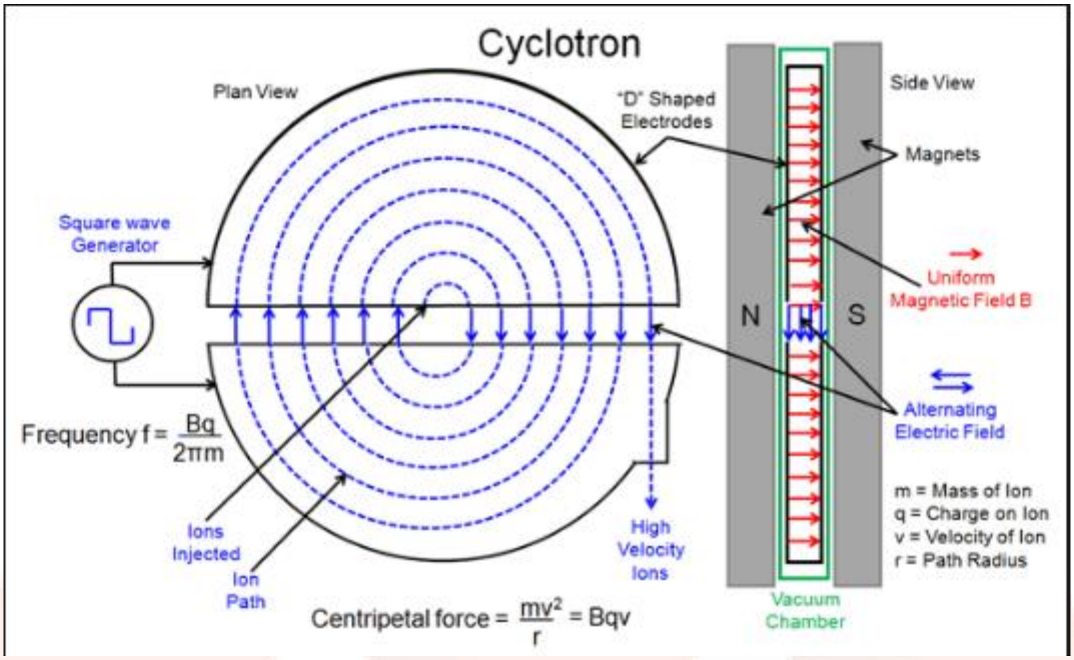
$r = \sqrt{y^2+a^2}$ dB= $\dfrac{\mu_0 Idy}{4\pi\left(y^2+a^2\right)^{3/2}}$

$$B = \frac{\mu_0 Idy}{4\pi}\int_{y=\infty}^{y=\infty}\frac{dy}{\left(y^2+a^2\right)^{3/2}}$$



b) **Write the working of cyclotron in brief. Draw a schematic sketch of the cyclotron showing path of accelerated charged particles (ions) in both dees.**
Derive expression for cyclotron frequency.

Sol.
Cyclotron brief: when positively charge particle or ions is made to move in a region of mutually $\perp^{ar}$ strong magnetic field and high frequency electric field time and again then it get accelerated aquire sufficiently large amount of K.E



Cyclotron Frequency :

$$r = \frac{mv}{qB}; \quad v = \frac{qBr}{m}$$

$$v = \frac{2\pi m}{qB} = \frac{qBr}{m}$$

$$T = \frac{2\pi m}{qB}$$

$$f = \frac{qB}{2\pi m}$$

$$w = 2\pi f = \frac{qB}{m}$$

30

30. (a) Derive an expression of the mirror equation, Draw the necessary ray diagram.

Sol:

$AP = -u,$

$A'P = -v,$

$FP = -f,$

$CP = 2FP = -2f$

From eqn. 3

$$\Rightarrow \frac{-2f-(-v)}{(-u)-(-2f)} = \frac{(-v)-(-f)}{(-f)}$$

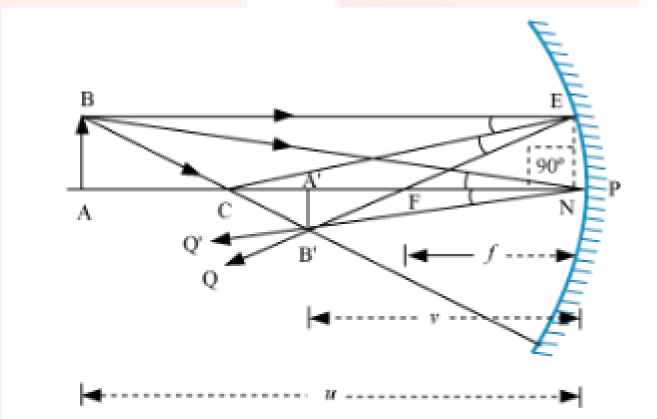
$$\Rightarrow \frac{v-2f}{2f-u} = \frac{v-f}{f}$$

$$\Rightarrow vf - 2f^2 = 2fv - uv - 2f^2 + uf$$

$$\Rightarrow uv = fv + uf$$

Dividing by uvf on both sides

$$\Rightarrow \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$



(b) Find the focal length of circle whose the radius of curvature is 10 cm.

Sol.

$R = 2f$

$f = \frac{R}{2}$

$f = \frac{10}{2} = 5cm$